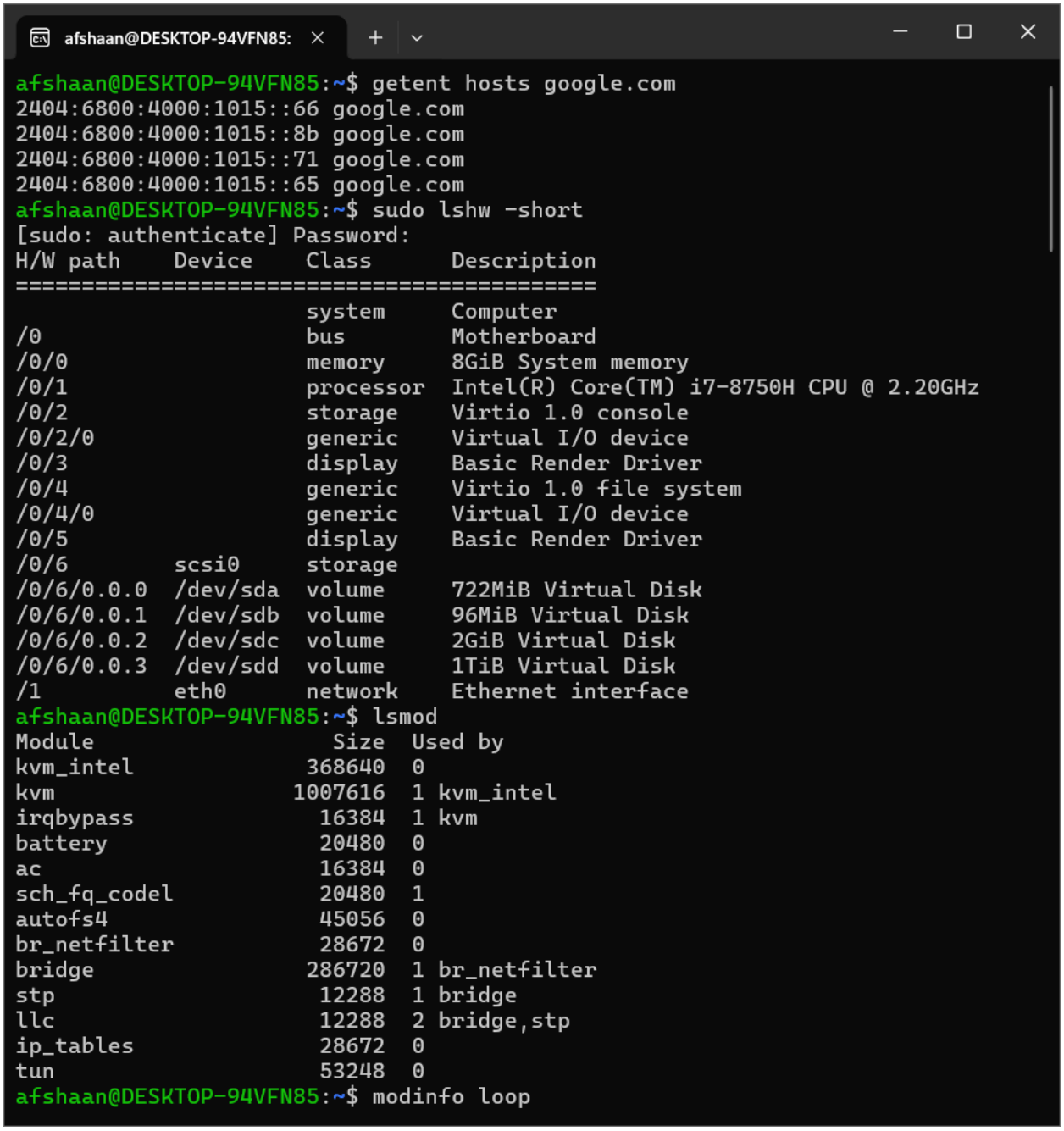
10 Linux Commands

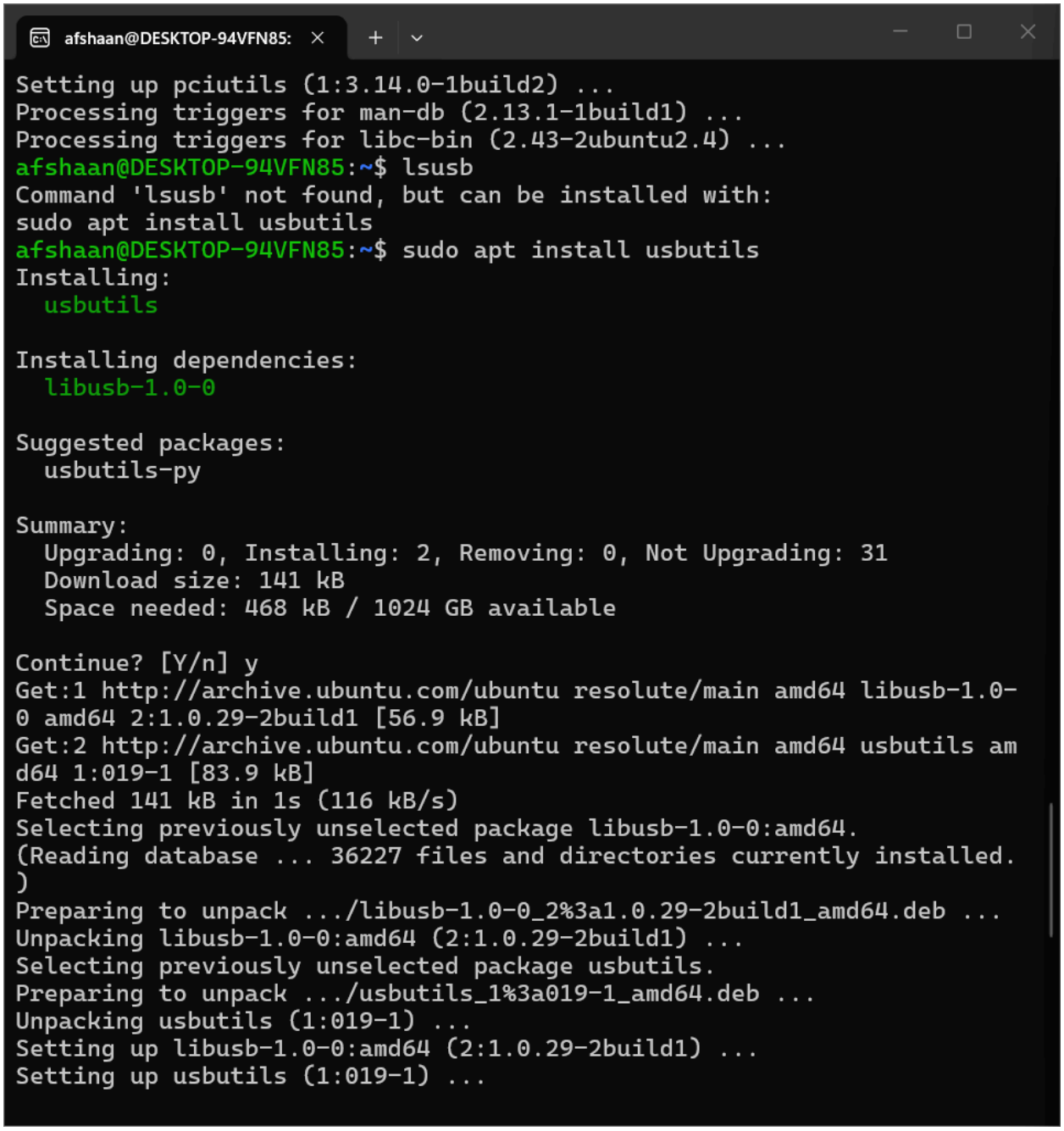
No.	Command	What it does	Practice example
1	getent	Retrieves information from system databases. (Previously explored; skip if familiar.)	getent hosts google.com
2	lshw	Displays hardware information.	sudo lshw -short
3	lsmod	Lists currently loaded kernel modules.	lsmod
4	modinfo	Displays details about a kernel module.	modinfo loop
5	dmesg	Displays kernel messages and hardware-related events.	dmesg --level=err,warn
6	sysctl	Displays or manages kernel parameters.	sysctl kernel.hostname
7	mkswap	Prepares a file or partition for use as swap space.	mkswap --help
8	blkdiscard	Discards unused blocks on supported storage devices.	blkdiscard --help
9	lspci	Lists PCI devices connected to the system.	lspci
10	lsusb	Lists USB devices connected to the system.	lsusb

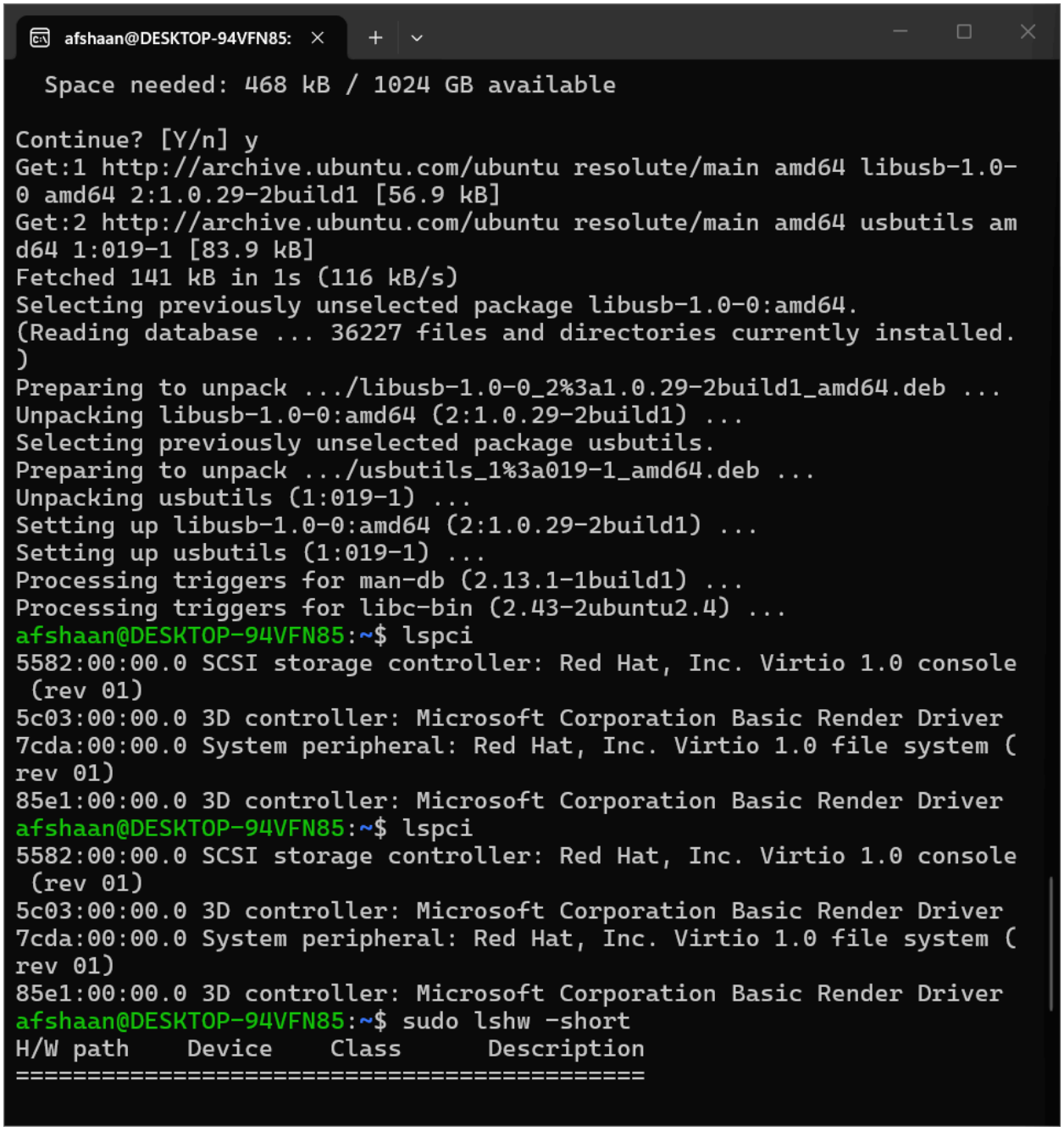

afshaan@DESKTOP-94VFN85:
afshaan@DESKTOP-94VFN85:~$ getent hosts google.com
2404:6800:4000:1015::66 google.com
2404:6800:4000:1015::8b google.com
2404:6800:4000:1015::71 google.com
2404:6800:4000:1015::65 google.com
afshaan@DESKTOP-94VFN85:~$ sudo lshw -short
[sudo: authenticate] Password:
H/W path Device Class Description
==========
system Computer
/0 bus Motherboard
/0/0 memory 8GiB System memory
/0/1 processor Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz
/0/2 storage Virtio 1.0 console
/0/2/0 generic Virtual I/O device
/0/3 display Basic Render Driver
/0/4 generic Virtio 1.0 file system
/0/4/0 generic Virtual I/O device
/0/5 display Basic Render Driver
/0/6 scsi0 storage
/0/6/0.0.0 /dev/sda volume 722MiB Virtual Disk
/0/6/0.0.1 /dev/sdb volume 96MiB Virtual Disk
/0/6/0.0.2 /dev/sdc volume 2GiB Virtual Disk
/0/6/0.0.3 /dev/sdd volume 1TiB Virtual Disk
/1 eth0 network Ethernet interface
afshaan@DESKTOP-94VFN85:~$ lsmod
Module Size Used by
kvm_intel 368640 0
kvm 1007616 1 kvm_intel
irqbypass 16384 1 kvm
battery 20480 0
ac 16384 0
sch_fq_codel 20480 1
autofs4 45056 0
br_netfilter 28672 0
bridge 286720 1 br_netfilter
stp 12288 1 bridge
llc 12288 2 bridge,stp
ip_tables 28672 0
tun 53248 0
afshaan@DESKTOP-94VFN85:~$ modinfo loop


afshaan@DESKTOP-94VFN85:
parm: max_loop:Maximum number of loop devices
parm: max_part:Maximum number of partitions per loop device (int)
parm: hw_queue_depth:Queue depth for each hardware queue. Default: 128
afshaan@DESKTOP-94VFN85:~$ dmesg --level=err,warn
[0.596033] MDS CPU bug present and SMT on, data leak possible. See https://www.kernel.org/doc/html/latest/admin-guide/hw-vuln/mds.html for more details.
[0.596033] MMIO Stale Data CPU bug present and SMT on, data leak possible. See https://www.kernel.org/doc/html/latest/admin-guide/hw-vuln/processor_mmio_stale_data.html for more details.
[1.029017] PCI: Fatal: No config space access function found
[1.068148] PCI: System does not support PCI
[1.490961] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
[3.506044] WSL (2 - init-systemd(Ubuntu)) WARNING: /usr/share/zoneinfo/Asia/Calcutta not found. Is the tzdata package installed?
[3.737249] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[3.743748] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.745162] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.746346] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.747687] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[3.751031] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.752022] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.753132] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.754098] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[3.754986] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[3.882211] Failed to connect to system scope bus via local transport: No such file or directory
[6.881033] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[7.307354] WSL (156) ERROR: CheckConnection: getaddrinfo() failed: -5
[6179.131207] WSL (156) ERROR: CheckConnection: getaddrinfo() failed: -5
[7618.275517] WSL (156) ERROR: CheckConnection: getaddrinfo() failed: -5
afshaan@DESKTOP-94VFN85:~$ sysctl kernel.hostname
kernel.hostname = DESKTOP-94VFN85
afshaan@DESKTOP-94VFN85:~$ mkswap --help
Usage:
mkswap [options] device [size]


afshaan@DESKTOP-94VFN85:
-5
[6179.131207] WSL (156) ERROR: CheckConnection: getaddrinfo() failed:
-5
[7618.275517] WSL (156) ERROR: CheckConnection: getaddrinfo() failed:
-5
afshaan@DESKTOP-94VFN85:~$ sysctl kernel.hostname
kernel.hostname = DESKTOP-94VFN85
afshaan@DESKTOP-94VFN85:~$ mkswap --help
Usage:
mkswap [options] device [size]
Set up a Linux swap area.
Options:
-c, --check check bad blocks before creating the swap a
rea
-f, --force allow swap size area be larger than device
-q, --quiet suppress output and warning messages
-p, --pagesize SIZE specify page size in bytes
-L, --label LABEL specify label
-v, --swapversion NUM specify swap-space version number
-U, --uuid UUID specify the uuid to use
-e, --endianness=<value> specify the endianness to use (native, litt
le or big)
-o, --offset OFFSET specify the offset in the device
-s, --size SIZE specify the size of a swap file in bytes
-F, --file create a swap file
--verbose verbose output
--lock[=<mode>] use exclusive device lock (yes, no or nonbl
ock)
-h, --help display this help
-V, --version display version
For more details see mkswap(8).
afshaan@DESKTOP-94VFN85:~$ blkdiscard --help


afshaan@DESKTOP-94VFN85:
For more details see mkswap(8).
afshaan@DESKTOP-94VFN85:~$ blkdiscard --help
Usage:
blkdiscard [options] <device>
Discard the content of sectors on a device.
Options:
-f, --force disable all checking
-l, --length <num> length of bytes to discard from the offset
-o, --offset <num> offset in bytes to discard from
-p, --step <num> size of the discard iterations within the offset
-q, --quiet suppress warning messages
-s, --secure perform secure discard
-v, --verbose print aligned length and offset
-z, --zeroout zero-fill rather than discard
-h, --help display this help
-V, --version display version
Arguments:
Values for <num> may be followed by a suffix: KiB, MiB,
GiB, TiB, PiB, EiB, ZiB, or YiB (where the "iB" is optional).
For more details see blkdiscard(8).
afshaan@DESKTOP-94VFN85:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
afshaan@DESKTOP-94VFN85:~$ sudo apt install pciutils
Installing:
pciutils
Installing dependencies:
libpci3
Suggested packages:
bzip2
Summary:


afshaan@DESKTOP-94VFN85:
Setting up pciutils (1:3.14.0-1build2) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
afshaan@DESKTOP-94VFN85:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
afshaan@DESKTOP-94VFN85:~$ sudo apt install usbutils
Installing:
usbutils
Installing dependencies:
libusb-1.0-0
Suggested packages:
usbutils-py
Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 31
Download size: 141 kB
Space needed: 468 kB / 1024 GB available
Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libusb-1.0-0 amd64 2:1.0.29-2build1 [56.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 usbutils amd64 1:019-1 [83.9 kB]
Fetched 141 kB in 1s (116 kB/s)
Selecting previously unselected package libusb-1.0-0:amd64.
(Reading database ... 36227 files and directories currently installed.)
Preparing to unpack .../libusb-1.0-0_2%3a1.0.29-2build1_amd64.deb ...
Unpacking libusb-1.0-0:amd64 (2:1.0.29-2build1) ...
Selecting previously unselected package usbutils.
Preparing to unpack .../usbutils_1%3a019-1_amd64.deb ...
Unpacking usbutils (1:019-1) ...
Setting up libusb-1.0-0:amd64 (2:1.0.29-2build1) ...
Setting up usbutils (1:019-1) ...


afshaan@DESKTOP-94VFN85:
Space needed: 468 kB / 1024 GB available
Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libusb-1.0-0 amd64 2:1.0.29-2build1 [56.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 usbutils amd64 1:019-1 [83.9 kB]
Fetched 141 kB in 1s (116 kB/s)
Selecting previously unselected package libusb-1.0-0:amd64.
(Reading database ... 36227 files and directories currently installed.)
Preparing to unpack .../libusb-1.0-0_2%3a1.0.29-2build1_amd64.deb ...
Unpacking libusb-1.0-0:amd64 (2:1.0.29-2build1) ...
Selecting previously unselected package usbutils.
Preparing to unpack .../usbutils_1%3a019-1_amd64.deb ...
Unpacking usbutils (1:019-1) ...
Setting up libusb-1.0-0:amd64 (2:1.0.29-2build1) ...
Setting up usbutils (1:019-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
afshaan@DESKTOP-94VFN85:~$ lspci
5582:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
5c03:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
7cda:00:00.0 System peripheral: Red Hat, Inc. Virtio 1.0 file system (rev 01)
85e1:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
afshaan@DESKTOP-94VFN85:~$ lspci
5582:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
5c03:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
7cda:00:00.0 System peripheral: Red Hat, Inc. Virtio 1.0 file system (rev 01)
85e1:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
afshaan@DESKTOP-94VFN85:~$ sudo lshw -short
H/W path Device Class Description